Junjie Yang

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EDUCATION BACKGROUND

South China University of Technology (SCUT) Sep 2022 - Jul 2026

- **Major:** Computer Science and Technology (Innovation Class)
- **Degree:** Bachelor of Science in Computer Science and Technology Expected in July 2026
- **Main Courses:** Artificial Intelligence, Neural Networks and Deep Learning, Data Structure, Computer Networks, Operating System, Introduction to Pattern Recognition, Data Warehouse and Data Mining, Software Engineering, Java Programming, Probability theory and mathematical statistics, Discrete Mathematics

PUBLICATIONS

Junjie Yang, MedGEN-Bench: Contextually Entangled Benchmark for Open-ended Multimodal Medical Generation, under submission to *The IEEE Conference on Computer Vision and Pattern Recognition,2026, website:* https://arxiv.org/abs/2511.13135

Wenxuan Shen[1], **Junjie Yang[1]**, Dongping Chen[2,3], Yaochen Wang[2], Mingjia Wang[2], Yao Wan[2], Weiwei Lin[1], InterFusion: Interleaved Embedding for Complex Document Retrieval, *under submission to The IEEE Conference on Computer Vision and Pattern Recognition,2026*

Wenxuan Shen, Mingjia Wang, Yaochen Wang, Dongping Chen, **Junjie Yang**, Yao Wan, Weiwei Lin, Are We on the Right Way for Assessing Document Retrieval-Augmented Generation? *website:* https://arxiv.org/abs/2508.03644

Shengsheng Lin[1], Weiwei Lin[1 2], Wentai Wu[3], Haojun Chen[1], **Junjie Yang[1]**, SparseTSF: Modeling Long-term Time Series Forecasting with 1k Parameters, *ICML 2024, website:* https://github.com/lss-1138/SparseTSF

Junjie Yang[1], Weiwei Lin[1 2], Shengsheng Lin[1], Wenxuan Shen[1], Feiyu Zhao[1], TSENet: Time Series Forecasting with Channel Squeeze-and-Excitation Network, *manuscript submitted to IEEE Transactions on Knowledge and Data Engineering (TKDE)*

INTERNSHIP

Shenzhen Research Institute of Big Data, The Chinese University of Hong Kong, Shenzhen Sep 2025 - Dec 2025

Position: *Undergraduate Research Lead & First Author* ***Advisor:*** *A.P. Changmiao Wang*

➢ **Project: MedGEN-Bench:Contextually Entangled Benchmark for Open-ended Multimodal Medical Generation**

- Led the entire project lifecycle as the first author, overseeing idea conceptualization, experimental design, code implementation, and cross-team coordination with clinicians and researchers; led the writing of the research paper, currently under submission to CVPR 2026.
- Developed a comprehensive multimodal benchmark containing 6,422 expert-validated image-text pairs, covering 6 imaging modalities, 16 clinical tasks, and 3 task formats.
- Designed a novel three-tier evaluation framework integrating pixel-level metrics, semantic text analysis, and expert-guided clinical scoring to assess model performance systematically.
- Conducted evaluations on 18 state-of-the-art models, revealing that compositional frameworks outperform unified models in cross-modal consistency.
- Optimized the medical expert review process by developing an intuitive data annotation website, enhancing the efficiency of benchmark validation through iterative clinician feedback.
- Demonstrated that contextual augmentation improves text-image semantic alignment by 36.3%, highlighting its critical role in clinical relevance.

Guangzhou Institute of Standardization Mar 2025 - Jun 2025

Position: *AI Algorithm Engineer of the Standardization Strategy and Quality Research Department*

➢ **Responsibilities:** Participated in two projects, mainly responsible for developing and optimizing NLP-related algorithms, and assisting the department in tasks including RAG system development, model training and optimization, as well as business application development.

➢ **Project 1: Automatic Review System of Bidding/Tendering Clauses Based on LLMs and Knowledge Enhancement**

- Worked as the lead developer in core architecture design, key algorithm implementation/optimization, and addressing data scarcity and insufficient model professionalism.
- Designed a Q&A data pipeline to generate question-answering pairs by GPT-4, Claude, and Deepseek and verify and filter data based on Self-Consistency, Self-Refine and few-shot expert evaluation.
- Built a vector knowledge base via LightRAG on high-quality legal Q&A data and used knowledge graph retrieval to align with legal experts' preferences.
- Participated in test scheme design and constructed large-scale test sets with normal/unfair clauses to support comprehensive model evaluation and deployment.
- Achieved >95% retrieval and ~60% accuracy on noisy test clauses and outperformed 4 intern models with top F1 score; adopted by Guangzhou Market Supervision Bureau.

➢ **Project 2: Unsupervised Classification of Complaint Work Orders Based on Hybrid Vector Representation and Hierarchical Clustering**

- Tackled unlabeled complaint classification and trend analysis in vague, colloquial user descriptions.
- Vectorized object information using text embedding models; calculated the distance matrix by weighting dense vectors and sparse vectors to achieve the matching of semantics and word frequency; implemented Hierarchical Density-Based Spatial Clustering of Applications with Noise on word vectors for unsupervised classification.
- Construct a semantic filtering pipeline: leverage large language models to analyze customer complaint descriptions and extract core complaint entities.
- Hybrid vector representation of entities: employ a weighted combination of dense and sparse vectors to achieve simultaneous

semantic matching and term frequency-based matching.

RESEARCH EXPERIENCES

ONE Lab, SCUT & HUST Mar 2025 - Sep 2025

Position: *Undergraduate Research Assistant* ***Advisor:*** *Prof. Weiwei Lin, Prof. Yao Wan*

➢ **Project 1: InterFusion: Interleaved Embedding for Complex Document Retrieval**

- Proposed InterFusion, a zero-shot multimodal Document RAG framework.
- Built InterFusion-400K, a large-scale, high-quality dataset by filtering ambiguous queries, generating captions, and constructing hard negatives from CoSyn-400K.
- Developed and trained InterQwen, an interleaved embedding model fine-tuned on InterFusion-400K.
- Tested on benchmarks using metrics like Hit@k, nDCG@k, and answer accuracy, and compared with SOTA text/visual embedding models and Document RAG frameworks.
- Demonstrated that InterQwen significantly outperformed existing embedding models and deliberately designed document RAG frameworks across benchmarks, with 39.3% and 3.5% Hit@1 improvement.
- Demonstrated that InterQwen achieved 5.9% and 3.3% improvement in answering accuracy compared with the previous state-of-the-art.

➢ **Project 2: Are We on the Right Way for Assessing Document Retrieval-Augmented Generation?**

- Created DOUBLE-Bench, the first large-scale, multilingual, and multimodal evaluation benchmark specifically designed for document RAG.
- Included over 62,000 document pages and generated 4,000 single-hop and multi-hop queries through a knowledge graph-driven process which underwent detailed annotation and expert review.
- Demonstrated that using multimodal embeddings yielded a 15.48% performance improvement in retrieval tasks compared to text-only embeddings and conducted in-depth analysis on the reasons for the improvement as well as the common bottlenecks in open-source multimodal document RAG frameworks.

Advanced Computing Architecture Team, SCUT Aug 2024 - Jun 2025

Position: *System Developer* ***Advisor:*** *Prof. Weiwei Lin*

➢ **Project 1: AI-powered Q&A Assistant for Local Chronicles** Nov 2024 - Mar 2025

- Participated in the development of an AI-powered historical assistant based on vertical applications of large language models, with Retrieval-Augmented Generation (RAG) as the core to enhance the efficiency of information retrieval and accuracy of responses in local chronicles.
- **Designed and implemented Iterative Retrieval Agent (IterRAG):** constructed a multi-turn retrieval-validation mechanism that autonomously evaluated answer quality, rewrote underperforming queries, and decomposed complex questions into iterative sub-queriesto, and used Auto-RAG which enabled LLMs to manage query decomposition, context maintenance, and final synthesis.
- Built user-centric front-end interface: adapted based on the open-source CherryStudio framework and provided functions such as user interaction, reference traceability, and knowledge graph visualization.
- Knowledge graph visualization: implemented Neo4j-based visualization to display the entity and relationships; enabled 2D/3D switching, zoom controls, and highlighting entities.

➢ **Project 2: SparseTSF: Modeling Long-term Time Series Forecasting with 1k Parameters** Aug 2024 - Jun 2024

- Introduced SparseTSF, a novel, extremely lightweight model for Long-term Time Series Forecasting (LTSF), which was designed to address the challenges of modeling complex temporal dependencies over extended horizons with minimal computational resources.
- Contributed to ablation study design and analysis for core model with key responsibilities including: wrote the ablation experiment section in research paper which was accepted as an ICML 2024 Oral Presentation, visualized model prediction performance on Electricity/Traffic datasets using Matplotlib, assisted in Linux server deployment, and implemented sparse neural network with only 1k parameters with PyTorch.

➢ **Project 3: TSENet: Time Series Forecasting with Channel Squeeze-and-Excitation Network** Sep 2024 - Jan 2025

- Independently developed a lightweight transformer-based architecture addressing critical limitations in multivariate time series forecasting: insufficient inter-channel correlation modeling and high self-attention complexity.
- Proposed a Channel Squeeze-and-Excitation Block, specifically designed to explicitly model channel relationships in a compact representation space.
- Used Time Channel Attention with linear complexity to generate global channel descriptors that responded to embedded channel features, allowing for adaptive recalibration of feature relationships between channels.
- Showed that TSENet had reached highest rank on forecast accuracy and outperformed other state-of-the-art models on 6 mainstream datasets.
- Completed a research paper, currently under review at IEEE Transactions on Knowledge and Data Engineering (TKDE, SCI Q1).

EXTRACURRICULAR ACTIVITIES & HONORS

The First Prize, Undergraduate Group, Asia and Pacific Mathematical Contest in Modeling 2023

The Second Prize, Undergraduate Group, the 13th MathorCup University Mathematical Modeling Challenge 2023

Honorable Mention, Interdisciplinary Contest in Modeling 2023

Technical Excellence Award (outperformed hundreds of participating teams), the 1st Guangdong-Hong Kong-Macau Greater Bay Area Data Application Innovation Competition 2023

PROFESSIONAL SKILLS

Computer Skills: Python, SPSS, MATLAB, C++

Languages: Native in Chinese (Mandarin), Fluent in English (TOEFL: 101, R 27/L 25/S 25/W 24)

Hobbies: Fitness Training (6 yrs.)